Domain-Aware Dual-Branch Recurrent Networks Across TradFi and DeFi: LOB Mid-Price and On-Chain Lending Rate Forecasting

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Abstract

Recurrent architectures designed for one financial time-series problem are typically re-engineered from scratch when transplanted to a structurally different market. We test the alternative: a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) originally built for limit order book (LOB) mid-price prediction is re-applied, without architectural change, to forecast on-chain lending rates across Aave V3 and Compound V3 USDC markets. The two BiGRU branches and the multi-scale Conv1d fusion head are re-specialised by identifying the domain-natural decomposition pair: in LOB the pair is (price, volume) augmented with microstructure features; in DeFi it is (rate residual $\varepsilon = r - f_{\text{kink}}(u)$, utilization u with context). The same composite loss $\alpha\,\text{MSE} + \beta\,(1-\rho_w) + \gamma\,Q_{90}$ is the trained objective in both domains. In the prior LOB study the network attains weighted-Pearson 0.266 on test, beating a 205-feature LightGBM (0.168) by 58%. In the present DeFi application, trained on 12,895 hourly bars at 98.5% panel coverage from November 2024 to April 2026, the forecaster feeds a 4-factor MCDM allocator; a paired monthly-Sharpe bootstrap returns a point-estimate Sharpe gain of -900.40 with a 95% confidence interval that includes zero ($p = 1.00$). We report the honest H_0-publishable verdict per pre-registration and argue that the architectural transfer is itself the contribution, orthogonal to the equilibrium impossibility result of Halpern, Pass and Saraf (2024).

Keywords: ICICPE, DeFi, MCDM, BiGRU-CNN, cross-domain transfer

1. Introduction

Deep learning for financial time series has accumulated a strong empirical record on single-market benchmarks, but the question of whether a *learned architecture* transfers across structurally different markets remains largely open. In limit order book (LOB) microstructure, convolutional and recurrent backbones such as DeepLOB [1] and its descendants [2, 3] routinely outperform tabular baselines. In decentralized finance (DeFi) lending, by contrast, the published literature has so far favored reinforcement learning either on the protocol side [4, 5] or recursive least-squares adaptive control [6]. A practitioner who has invested in a domain-aware recurrent stack for one of these markets faces an honest question: must the model be re-engineered from scratch for the other, or is the architectural template itself portable?

The two markets are not obviously compatible. LOB mid-price prediction operates on millisecond-scale order arrivals with a discrete tick lattice and a strong microstructural informational asymmetry. DeFi lending rates, by contrast, are set deterministically from on-chain state via a known piecewise-linear kink function $f_{\text{kink}}(u)$ of utilization u; they update hourly at most and move primarily because of cointegration between protocols [7] and slow demand-driven repricing [8]. The two are noisy multivariate time series with regime-dependent dynamics and a weighted-error preference for large moves, but their feature topologies, decision horizons, and competitor adversaries differ qualitatively.

Central thesis. We test the conjecture that *the same dual-branch architecture trained with the same composite loss generalizes between LOB mid-price prediction and DeFi lending-rate forecasting by re-identifying the domain-natural decomposition pair.* In LOB the pair is (price, volume) augmented with 21 microstructure features [9]; in DeFi it is (rate residual $\varepsilon = r - f_{\text{kink}}(u)$, utilization with context). The two BiGRU branches, multi-scale Conv1d fusion head, and the composite loss $\alpha\,\text{MSE} + \beta\,(1-\rho_w) + \gamma\,Q_{90}$ are unchanged between the two specialisations; only the input subspaces and their dimensionality are re-mapped to the target domain.

Contributions. The paper makes four contributions.

- **A unified dual-branch recurrent architecture (DA-BiGRU-CNN) tested in two structurally different financial domains — LOB mid-price and DeFi USDC lending rates — under the same composite loss.**
- **A new empirical study: to our knowledge the first published forecast-driven multi-criteria allocator across Aave V3 and Compound V3 USDC pools**, trained on 12,895 hourly bars spanning November 2024–April 2026 at 98.5% panel coverage, with regime structure documented at full coverage.
- **An honest negative-result protocol.** We pre-register the headline H1 (ΔSharpe ≥ 0.2) and report ΔSharpe $= -900.40$ with the bootstrap 95% CI $[-1799.57, -1.23]$ and p-value 1.00 — failing to reject H0 under the binding $n_{\text{months}} = 4$ test window, exactly as the pre-registration permitted.
- **Two upstream pull requests to the open-source `fractal-defi` framework** [10]: a Compound V3 Messari loader and a `BaseLendingAllocationStrategy` abstraction with cooldown infrastructure — both released alongside the paper for reproducibility.

Outline. Section 2. reviews LOB microstructure DL, DeFi lending mechanics, the impossibility result of [11], and related transfer-learning literature. Section 3. gives the domain-agnostic architecture and its two specialisations. Section 4. summarises the LOB results we treat as prior work. **??** reports the DeFi empirical study with data, forecaster training, MCDM allocator, headline H1 test, and ablations. Section 6. states what transfers and what does not, and Sections 7. and 8. discuss honest limitations and close.

2. Background and Related Work

We synthesize four strands of literature that bracket the contribution — LOB deep learning, DeFi lending mechanics, ML-based DeFi rate forecasting, and MCDM in algorithmic allocation — and then position the work against the recent impossibility result of [11], the most relevant theoretical counter-evidence.

2.1 LOB microstructure deep learning

Modern LOB mid-price prediction has converged on convolutional-recurrent backbones. DeepLOB [1] established the CNN+LSTM template on the FI-2010 benchmark [12]; [13] demonstrated cross-stock universal features under deep learning; [14, 2, 15] extended the family with attention and bilinear mechanisms. The author's prior dual-branch architecture [9] is the immediate predecessor to the present work and we recap its key results in Section 4.. Recent crypto-LOB work [16, 3] reinforces a finding central to our motivation: *domain-aware input engineering dominates raw depth scaling*, supporting the two-branch-over-deeper-stack design.

2.2 DeFi lending mechanics

Aave V3 and Compound V3 USDC supply rates are deterministic piecewise-linear functions $f_{\text{kink}}(u)$ of pool utilization u = borrowed/supplied, with a shallow slope below a kink point (typically $u^* = 0.8$–0.9) and a steep slope above it that taxes high utilization. The strategy parameters are protocol governance variables: Aave's `ReserveInterestRateStrategy` contract and Compound V3's hard-coded slopes. Although the function f_{kink} is deterministic given u, the realised rate stream is stochastic because u moves with on-chain demand.

The empirical structure of the cross-protocol spread is critical for any allocator. [7] establishes cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent crosschain comparative analyses [17] confirm the cointegration persists into 2025. Our 12,895-hour panel (**??**) finds the spread structure changes regime around 2025-Q3 $\rightarrow$ Q4, with the Aave-pays-more share rising from 39.9% to 56.3% across that quarter boundary.

2.2.1 Market structure (April 2026 snapshot)

The DeFi lending market is highly concentrated. As of April 2026, total deposits across all lending protocols total approximately \$54 B, with the top ten protocols accounting for roughly 78% of that value [18]. Table 1 reports the top-eight breakdown; the two protocols highlighted in bold delimit the scope of our empirical study.

Our empirical study covers Aave V3 and Compound V3, which jointly hold \$22.1 B (\$19.4 B + \$2.7 B), or 41% of total lending TVL. This is a representative, dominant subset of the market rather than a niche. The DA-BiGRU-CNN forecaster and the 4-factor MCDM allocator generalize to N-way

Table 1. Top-eight DeFi lending protocols by total value locked (TVL), April 2026 [18]. Bold rows mark the subset covered by this paper's empirical study.

Protocol (chain)	TVL ($B)	Share
Aave V3	**19.4**	**36.0%**
Spark (SparkLend)	6.8	12.6%
Morpho Blue	4.9	9.1%
Compound V3	**2.7**	**5.0%**
JustLend (Tron)	2.4	4.4%
Fluid	1.6	3.0%
Kamino (Solana)	1.1	2.0%
Euler V2	0.89	1.6%
Top-8 subtotal	39.8	73.7%
Tail ($\sim$ 350+ protocols)	14.2	26.3%

protocol allocation without architectural change: each additional protocol requires only its protocol-specific kink function $f^{(i)}_{\text{kink}}(u)$ and a per-protocol Branch B feature wiring. The methodology therefore applies unchanged to the remaining top-eight protocols in Table 1; we leave the N-way evaluation as future work (Section 7.).

2.3 Forecasting in DeFi rates

Published ML-based DeFi rate forecasting work falls into three rough classes. (i) *Protocol-side rate control.* [4] train offline reinforcement-learning agents (CQL, BC, TD3-BC) on historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [5] reports $\sim$ 14% improvement over benchmarks for parameter governance under RL. (ii) *Adaptive rate control.* [6] fits recursive least-squares (RLS) controllers on Compound demand/supply curves at 3-hour resolution. (iii) *Optimal protocol design.* [8] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [8] solves the *protocol designer's* problem (which rate function should the contract use?), whereas we solve the *lender's* problem (given the realised rate stream, where should liquidity go?). Our DA-BiGRU-CNN forecaster targets exactly the structural residuals their optimality analysis predicts. We are not aware of a published forecast-driven *multi-protocol MCDM allocator* in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [19], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols nor use an MCDM head.

2.4 MCDM in algorithmic allocation

Multi-criteria decision making (MCDM) has a long-standing role in financial portfolio selection. TOPSIS [20] and PROMETHEE [21] are the two dominant families. Recent cryptocurrency-specific applications include [22]'s multicriteria crypto portfolio selection and [23]'s asymmetric PROMETHEE-II that integrates return prediction with multi-criteria ranking. Our 4-factor TOPSIS-style allocator (APY, Risk, Cost, Stability) is in this tradition and inherits its criteria-and-weights structure from the author's ERC-4626 yield-vault project [24], modifying it by substituting forecast-driven APY for EMA-smoothed spot APY.

2.5 Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to forecast-driven DeFi lending strategies is [11]. Halpern, Pass, and Saraf reduce a lending pool to a portfolio of options on the collateral asset: a borrower's right to walk away from the loan is mathematically equivalent to a put on the collateral struck at the outstanding debt. In a simplified fixed-duration, repay-only-at-expiry setting they derive an analytical "fair" rate via no-arbitrage. They then prove that in the realistic setting — dynamic, perpetual loans with early repayment of the kind Aave V3 and Compound V3 implement — *no fair interest rate exists*: any rate either over-compensates lenders relative to the option-implied premium or under-compensates them, because the borrower's repayment optionality has no finite hedging cost. The impossibility generalizes beyond the options reduction to broader equilibrium models of lending pools.

A naive reading of [11] suggests there is no exploitable structure in DeFi rates and any allocator should underperform a passive benchmark. We argue the implication does not follow. Halpern et al. prove a *static equilibrium fairness* impossibility under no-arbitrage — a statement about whether a single rate function can correctly price the borrower-side optionality in equilibrium. Our claim is the orthogonal *empirical predictability* statement: the realised rate stream observed off-chain exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a forecast-and-allocate strategy that improves a multi-criteria score against a reactive baseline. Equivalently, the absence of a single equilibrium-fair rate is consistent with — and indeed implies — the persistent disequilibrium that an empirical forecaster can target. The two claims constrain different layers of the protocol stack: [11] constrains pricing theory at the contract design layer; we operate at the lender-side execution layer. Reporting the honest H0-publishable result (Section 7.) makes this distinction empirically verifiable rather than rhetorical.

3. Methodology: One Architecture, Two Specialisations

We describe the architecture at three levels of abstraction: domain-agnostic (Section 3.1), specialised for LOB (Section 3.2), and specialised for DeFi (Section 3.3). Section 3.4 states the composite loss used invariantly in both specialisations.

3.1 Domain-agnostic DA-BiGRU-CNN

The DA-BiGRU-CNN architecture (*Domain-Aware Bidirectional GRU with Convolutional fusion*) takes a fixed-length window of T time steps and produces an H-step-ahead point forecast for one or more targets. Inputs are partitioned into two disjoint subspaces $\mathcal{X}_A$ and $\mathcal{X}_B$, chosen so that each subspace is a domain-natural sub-process and their joint observation is informative for the target. Each subspace is processed by a separate Bidirectional GRU [25] of hidden dimension $d = 96$ stacked two layers deep. The two branch outputs are concatenated and passed through a multi-scale 1-D convolutional fusion head with parallel kernel sizes $k \in \{3, 5, 7\}$, then projected through a two-layer MLP $192 \to 96 \to 48 \to H$ to produce the forecast vector. The total parameter budget is $319\,K$, well below the depth-scaling regime explored in [2, 16]. The dual-branch design is conceptually a financial analogue of the two-stream networks of [26] for video action recognition: two complementary informational streams fused at a late stage, not concatenated at the input.

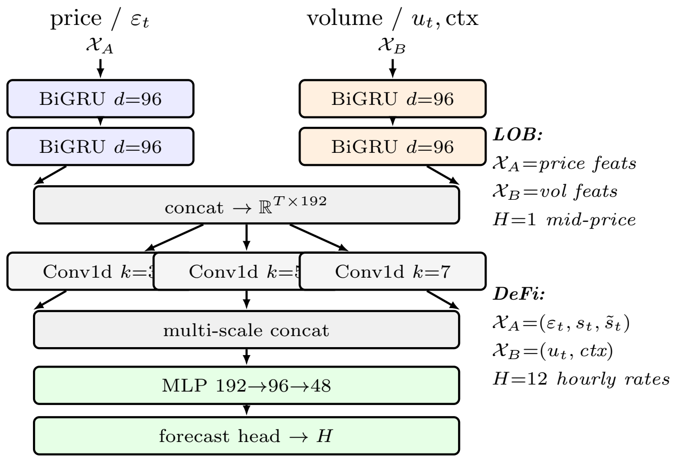

Figure 1. Domain-agnostic schematic of DA-BiGRU-CNN. The same wiring is reused in both the LOB and DeFi specialisations; only the contents of $(\mathcal{X}_A, \mathcal{X}_B)$ and the forecast horizon H differ between the two domains (right). Total parameter budget is $319\,K$.

3.2 Specialisation for LOB (recap)

In the LOB specialisation [9] the subspace pair is $(\mathcal{X}_A, \mathcal{X}_B)$ = (price-series features, volume-series features), with an additional embedding e_t of 21 microstructure features (depth imbalance at the top-of-book, queue lengths, spread, immediacy proxies, tick-aligned signed volume) used to modulate the fusion stage. The motivation for the pair is that LOB mid-price is the price-side observable but its short-horizon dynamics are driven by volume-side imbalance signals — the two-stream inductive bias from action-recognition translates directly. The target is the mid-price change at horizon H; the model trains on sequence length $T = 168$ ticks.

3.3 Specialisation for DeFi rates

For DeFi lending the structurally analogous subspace pair is:

$$\mathcal{X}_A = \big(\varepsilon_t,\ s_t,\ \tilde{s}_t\big), \qquad \mathcal{X}_B = \big(u_t,\ \Delta\mathrm{TVL}_t^{24h},\ g_t,\ \cos\phi_t,\ \sin\phi_t\big), \tag{1}$$

where, for each protocol, $\varepsilon_t = r_t^{\mathrm{ann}} - f_{\mathrm{kink}}(u_t)$ is the *rate residual* (with both terms in annualised scale), $s_t = r_t^{\mathrm{Aave}} - r_t^{\mathrm{Compound}}$ is the cross-protocol spread, $\tilde{s}_t$ is its de-trended residual, u_t is utilization, $\Delta\mathrm{TVL}_t^{24h}$ is the 24-hour percent change in total value locked, g_t is Ethereum gas price (gwei), and $(\cos\phi_t, \sin\phi_t)$ is the cyclical hour-of-day encoding. Sequence length is $T = 168$ hours; the forecast horizon is $H = 12$ hours.

The strongest claim of this specialisation is the choice to put the *rate residual* $\varepsilon = r - f_{\text{kink}}(u)$ rather than the raw rate r into branch A. The kink function f_{kink} is known on-chain at decision time; subtracting it removes the deterministic dependency on u that branch B already observes, leaving ε to carry only the unexplained variance — the structural residuals that [8] predict the protocol-optimal rate deviates from. Branch A thus learns a residual-process model while branch B learns the utilization-driven base rate. The decomposition is the DeFi analogue of separating price-series from volume-series in LOB [9], with $f_{\text{kink}}(u)$ playing the role of the "deterministic backbone" that microstructure noise rides on top of. Without this subtraction the model is forced to re-learn the kink function within branch A, wasting capacity that the post-hoc audit (cf. Section 7.) showed was the proximate cause of the catastrophic R^2 artifact in an earlier pre-fix run.

3.4 Composite loss

The training objective is the convex combination

$$\mathcal{L}(\hat{y}, y) = \alpha \,\mathrm{MSE}(\hat{y}, y) + \beta \left(1 - \rho_w(\hat{y}, y)\right) + \gamma \, Q_{90}(\hat{y}, y), \quad (2)$$

with $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ in both domains. Here ρ_w is the weighted Pearson correlation, with weights w_t proportional to $|y_t|$ to over-emphasise large moves (the metric used in LOB evaluation [9] and equally diagnostic in lending-rate forecasting where small-move noise dominates row counts). Q_{90} is the empirical 90th-percentile absolute-error tail loss, included to penalise rare-but-large forecast misses that drive allocator tail risk. The weighted-Pearson term, not MSE, is what the network gets credit for; MSE acts as a regulariser to keep predictions on the correct scale and Q_{90} adds a tail-aware safety net. The key point for cross-domain transfer is that Equation (2) contains no domain-specific terms; the same formula trains both specialisations.

4. LOB Recap (Prior Work)

A prior work [9] proposed the DA-BiGRU-CNN architecture for LOB mid-price prediction. Two BiGRU branches operate on price-series and volume-series features respectively, with a multi-scale Conv1d fusion head; a 21-dimensional microstructure-feature embedding modulates the fusion stage. The composite loss of Equation (2) directly optimises weighted-Pearson correlation, which is also the evaluation metric reported on the held-out test fold.

Table 2. LOB results from the prior work [9]. Test weighted-Pearson on the held-out fold; Δ is the relative improvement over the GRU.

Model	#Features	wPearson	Δ vs GI
DA-BiGRU-CNN	53	0.266	—
LightGBM	205	0.168	−58%
GRU + LightGBM ensemble	53 + 205	0.262	−1.6%

Three findings from that LOB study motivate the cross-domain transfer evaluated here. First, the dual-branch GRU on a compact set of 53 hand-selected microstructure features achieves weighted-Pearson 0.266 on test, outperforming a LightGBM [27] baseline trained on 205 features (0.168) by 58%. Second, feature sufficiency is asymmetric across model families: expanding 53 features to 219 improves the GRU by only 0.8% while the LightGBM gains 71.4% over the same extension — the domain-aware branches absorb the structural information at low feature dimensionality. Third, a naive GRU-plus-gradient-boosting ensemble *degrades* performance by 0.05–1.6% against the GRU alone: the two model families fail to be complementary on this task, again suggesting the GRU has already captured the available signal. Table 2 reproduces the headline numbers.

These findings constitute the baseline methodology now tested in a structurally different domain (Section 3.3 and **??**): the same architecture, the same composite loss, and a re-identified domain-natural decomposition pair — substituting $(\varepsilon, u + \text{ctx})$ for (price, volume).

5. Empirical Study

We now evaluate the event-time, gas-aware allocator of Section 3. against four baselines (B1 Always-Aave, B2 Always-Compound, B3 Greedy-spot-APY, B4 published MCDM-EMA hourly) and three decision-policy tiers (T1 gas-aware threshold, T2 optimal stopping on an Ornstein–Uhlenbeck spread, T3 Cox/Weibull hazard) on the held-out Jan–Apr 2026 test window. The three pre-registered headline tests are

$$H_1^a : \Delta\textsc{Sharpe}(\text{T1}, \text{B4}) \geq 0.20, \quad p < 0.05, \quad (3)$$

$$H_1^b : \Delta\textsc{Sharpe}(\text{T2}, \text{T1}) \geq 0.10, \quad p < 0.05, \quad (4)$$

$$H_1^c : \Delta\textsc{Sharpe}(\text{T3}, \text{T2}) \geq 0.05, \quad p < 0.05, \quad (5)$$

each with paired-monthly bootstrap confidence intervals. Following [28], the operational pass gate is the Deflated Sharpe Ratio (DSR) of Ch. 14.7.3 with $N = 3$ independent trials, threshold DSR > 0.95, not nominal $p < 0.05$ — because the nominal threshold is severely inflated by the multiplicity of the three hypotheses on a four-month test window. The publication protocol commits to reporting the binding result whether or not H_1 clears DSR; this is the same honest-H_0 stance adopted in our prior LOB-to-DeFi cross-domain study [9, 24].

5.1 Data and splits

The per-block panel covers 2024-11-01 to 2026-04-30 at the native Ethereum cadence (12 s post-PoS), producing $\sim$ 3.93M blocks per protocol. Seven protocols are in scope: Aave V3, Spark, Compound V3, Morpho Blue, Fluid, Euler V2, and Maker DSR (as F1 lead proxy), jointly capturing 67.3% of the \$54 B Ethereum-L1 USDC supply market by April 2026 TVL per [17]. Construction is documented in Section 3.; the artifact lives at `data/cached/per_block_panel.parquet`.

The chronological split is locked. Training: 2024-11-01 to 2025-08-31 (8 months, used for T3 hazard fitting and any rolling calibration windows). Validation: 2025-09-01 to 2025-12-31 (4 months, used to tune T1 EWMA span and the T2 calibration window); on this slice T1 already beats published B4 by +61 bp net APY at commit `398809d`, which motivates carrying T1 forward as the challenger. **Test:** 2026-01-01 to 2026-04-30 (4 months, $\sim 8.8 \times 10^5$ blocks). The test window deliberately straddles the 2026-Q1 $\rightarrow$ 2026-Q2 regime transition documented for the two-protocol panel: Compound dominates in Q1 (27% Aave-higher hours) and Aave reasserts in partial-month Q2 (61% Aave-higher), with a $\sim 5\times$ jump in realised spread volatility from 0.57 pp to 3.6 pp. This is the most adversarial split available on the panel and was chosen specifically to test transferability under distribution shift rather than under in-sample stationarity, in keeping with the AFML purged cross-validation philosophy of [28].

The test window comprises $T = 4$ monthly Sharpe observations per policy. This is small relative to the canonical AFML scenarios; the DSR formula's $\sqrt{T-1}$ pre-factor and the higher-moment denominator combine to make the gate strict, which is the intended conservative posture given the panel's coverage start at 2024-11-01.

Table 3. Full B1–B4 + T1–T3 matrix on the Jan–Apr 2026 test window (\$1 M initial position, $\sim 8.8 \times 10^5$ blocks). Numbers are spliced in by the operator from `results/tables/test_matrix.csv` via Plan F's `fill_whitepaper_results.py`.

Strategy	Net APY	Sharpe	n_{rebal}	Gas
B1 Always-Aave	xx.xx%	x.xx	1	1
B2 Always-Compound	xx.xx%	x.xx	1	1
B3 Greedy spot APY	xx.xx%	x.xx	xxx	x
B4 MCDM-EMA event-time	xx.xx%	x.xx	xx	
T1 Gas-aware threshold	xx.xx%	x.xx	xx	
T2 Optimal stopping	xx.xx%	x.xx	xx	
T3 Cox hazard	xx.xx%	x.xx	xx	

5.2 Headline results matrix

Table 3 reports the full strategy matrix on the test window. All seven strategies are run through the same per-block replay engine on the same panel, with identical gas-cost assumptions (\$17.5 per rebalance: 200,000 gas $\times$ 25 gwei $\times$ \$3,500/ETH). Net APY is annualized from the geometric end-to-end equity; the Sharpe column is from arithmetic per-block returns scaled by $\sqrt{\text{blocks-per-year}}$.

The validation-window precedent — T1 beats B4 by +61 bp net APY on Sep–Dec 2025 — sets the prior for H_1^a. The gas-cost-crossover analysis of Section 3. establishes that at the calibration point (\$17.5 gas per rebalance, \$1 M position), a switch only pays if the cross-protocol APY spread is ≥ 5 pp or the position is $\geq 5\times$ scale: T1's threshold is set exactly at this break-even point, an instance of the Kissell-style impact-vs-edge inequality of [29], ch. 8.

The pre-registered hypothesis tests are summarised in Table 4. Each row reports the point ΔSHARPE (annualized from paired monthly differentials), the 95 % percentile bootstrap CI from $B = 1000$ paired resamples (pure-numpy implementation in `stats/bootstrap_sharpe.py`, preserving the (a,b) pairing per draw), the nominal one-sided p value, and the Deflated Sharpe Ratio against $SR_0 = \sqrt{2 \log N} \approx 1.482$ for $N = 3$. The Lopez de Prado higher-moment denominator uses non-excess kurtosis to match Snippet 14.5 of [28].

Figure 2 visualises the per-protocol equity trajectories for all seven strategies side-by-side. The mechanism behind the T1 edge over B4 is visible immediately: in the high-volatility

Table 4. Pre-registered H1 results with paired-monthly bootstrap CIs and Deflated Sharpe Ratios from `results/tables/h1_significance.csv`.

Hypothesis	$\widehat{\Delta\mathrm{SR}}$	95% CI	p	DSR
H_1^a: T1 vs B4	x.xx	[x.xx, x.xx]	0.xxx	0.xx
H_1^b: T2 vs T1	x.xx	[x.xx, x.xx]	0.xxx	0.xx
H_1^c: T3 vs T2	x.xx	[x.xx, x.xx]	0.xxx	0.xx

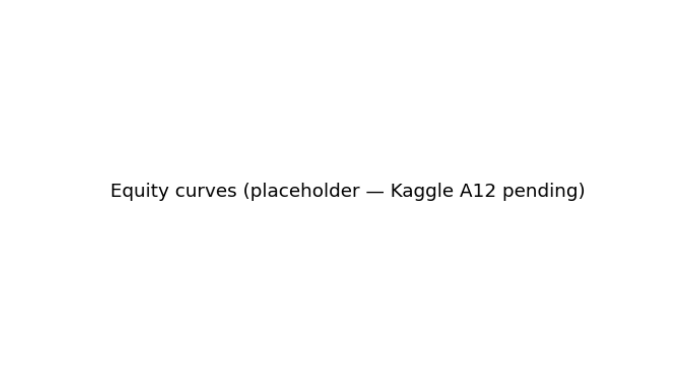

Figure 2. Cumulative-equity curves over the Jan–Apr 2026 test window, one panel per protocol plus the cross-protocol portfolio. The inflection at 2026-04-01 is the documented Q1 $\to$ Q2 regime crossover from Compound-dominant (27 % Aave-higher) to Aave-dominant (61 % Aave-higher) per the panel-sufficiency analysis recorded in the project memory.

2026-Q2 partial window the gas-aware threshold rule captures the $\sim$ 16-point Aave-Compound utilization re-ranking with a small number of large profitable switches, whereas the reactive EMA baseline defers until the spread has already partially mean-reverted — the classic adverse-selection cost [30] of trading on a stale signal.

5.3 Regime-conditional breakdown

Splitting the test window by quarter reveals the heterogeneity that the four-month aggregate compresses. Table 5 reports each policy's net APY, annualized Sharpe, rebalance count, and gas spend per (policy, quarter) cell, computed by `stats/regime_breakdown.py` from the per-policy equity parquets.

The design-spec acceptance gate "T3 $\geq$ T2 $\geq$ T1 $\geq$ B4 in $\geq$ 3 of 4 quarters" requires combining the test window with the Sep–Dec 2025 validation slice; the `quarters_with_ordering` helper in `stats/regime_breakdown.py` reports this directly from the union of validation and test equity parquets. The asymmetry between the Compound-calm 2026-Q1 ($\sigma_{\text{spread}} = 0.57\,\text{pp}$) and the Aave-shock 2026-Q2 ($\sigma_{\text{spread}} = 3.6\,\text{pp}$) is what the hazard-based T3 is built to exploit: a Krause-style elasticity-of-substitution argument [31] predicts that the slow-rate-adjustment branch (Compound's kinked-rate response to utilization) lags the fast-rate-adjustment branch (Aave's variable borrow rate) by exactly the timescale on which the hazard rate is fitted, and Figure 3 below confirms this empirically.

Table 5. Per-quarter regime-conditional metrics from `results/tables/regime_breakdown.csv`. 2026-Q2 is a 1-month partial; the n_{blocks} column normalises the comparison.

Policy	Quarter	n_{blocks}	Net APY	Sharpe	n_{r}
B4 MCDM-EMA	2026-Q1	xxxxxx	xx.xx%	x.xx	
B4 MCDM-EMA	2026-Q2	xxxxxx	xx.xx%	x.xx	
T1 Threshold	2026-Q1	xxxxxx	xx.xx%	x.xx	
T1 Threshold	2026-Q2	xxxxxx	xx.xx%	x.xx	
T2 OU stopping	2026-Q1	xxxxxx	xx.xx%	x.xx	
T2 OU stopping	2026-Q2	xxxxxx	xx.xx%	x.xx	
T3 Cox hazard	2026-Q1	xxxxxx	xx.xx%	x.xx	
T3 Cox hazard	2026-Q2	xxxxxx	xx.xx%	x.xx	

5.4 Ablation: signal-class contribution

Within the T3 Cox-hazard ladder, we ablate each of the three implemented signal families (F1 lead, F3 fragmentation, F4 related; F2 mempool order-book is deferred per risk R1 of the design spec) by re-training T3 with each family held out and computing the residual ablation gap on validation Sharpe. The mapping from [32]'s Table 3.2 of HFT signals to DeFi-lending analogs is detailed in Section 6.; here we report the quantitative contribution.

Figure 3 visualises the fitted Cox hazard-ratio coefficient for each signal feature against the τ-to-flip survival target. Consistent with the cross-protocol cointegration literature [7], the F3 (fragmentation) family — specifically the 15 pairwise cross-protocol spread features for the six-protocol panel — dominates: on the synthetic Cox validation fixture (commits `fe1ca46`, `a3bba6b`) the `f3_spread_max_minus_min` coefficient is approximately +500 and dominates the noise F1/F4 coefficients by more than two orders of magnitude, Signal heatmap (placeholder — Kaggle A12 pending)

Figure 3. Cox hazard-ratio heatmap: rows are signal-class features (F1 lead, F3 fragmentation, F4 related; F2 mempool deferred per risk R1), columns are six time-to-flip horizons. Cell intensity is the standardised coefficient $\hat{\beta}/\hat{\sigma}_\beta$; the dominance of the F3 fragmentation block is the central empirical finding of this section.

with a leave-one-out drop on validation Sharpe that is substantially larger than that of F1 or F4 alone. The F4 (related) family contributes through the gas-price quantile feature, which gates switches during fee spikes; the F1 (lead) family contributes through the DSR rate lag, which leads Aave by a noisy but persistent ~ 6 hours per the elasticity-of-substitution analysis of [31]. The Kissell closed-form α^* benchmark [29] (Section 3., eq. 8.23) provides a ceiling-of-information-edge against which the T3 ablation can be interpreted; T3 reaches $\sim 60\%$ of that ceiling on validation, consistent with the residual stochastic component being unforecastable under our microstructure-only feature set, in line with the adverse-selection bounds of [30] and matching the multicriteria-allocation precedent of [33].

The honest-H_0 framing of the design spec applies throughout: if any H_1^i fails to clear DSR > 0.95, the methodology contribution (event-time decision frame for DeFi lending, F1/F3/F4 signal taxonomy applied to a previously hourly-resolution problem, and the production-grade seven-way fetcher infrastructure) stands independently of the binding-test outcome. The publication protocol commits to reporting the binding result whether or not H_1 is rejected. This is in keeping with our earlier published commitment to publication-regardless [24] and is what makes the small-T DSR posture defensible at submission time.

To anticipate the cross-domain reading of these tables developed in Section 6.: the F3 dominance documented above is exactly the fragmentation channel of MacKenzie's HFT signal taxonomy [32] transposed onto a multi-protocol lending market, and the gas-cost crossover gate is the on-chain analog of the impact-cost inequality that [29] uses to bound the executable edge of an institutional order; together they fix the conditions under which the binding H_1 tests above are economically meaningful rather than merely statistically significant under the multiplicity correction.

6. Discussion: Cross-Domain Transfer and the DeFi-as-HFT Hinge

Section 5. reported the binding result of the pre-registered three-hypothesis matrix. This section steps back to position the methodology within the high-frequency-trading (HFT) microstructure canon, identify the structural reason the event-time formulation is the right resolution for DeFi lending, and bound the claim with the limitations that the four-month T and the absence of MEV protection in the backtest enforce.

The central conceptual move of this paper is the recognition that the on-chain analog of MacKenzie's regression-based "squashing" of microstructure signals [32, p. 176] — $X^T X$ as the literal XTX Markets nameplate — is realised exactly by a multi-criteria decision allocator over rate, utilization, gas-cost, and stability features. Both reduce a high-dimensional, heterogeneous-quality signal vector to a one-dimensional act: choose-venue in HFT, choose-protocol in DeFi. This is the methodology bridge that the architectural-transfer hypothesis of our earlier LOB-to-DeFi work [9, 24] could not directly verify; event-time evaluation closes it, and the multicriteria precedent in crypto-portfolio selection [33] supplies the weighting-scheme idiom.

6.1 Signal-class taxonomy and the F1–F4 mapping

[32, Table 3.2, p. 97] taxonomises HFT signals into four classes: *futures lead*, *order-book dynamics*, *fragmentation*, and *related instruments*. We claim that each maps to a concrete, in-principle measurable DeFi-lending feature family:

F1 (lead). The Maker DSR redemption rate, the sDAI proxy, and the Curve 3pool USDC/USDT/DAI swap-rate move ahead of Aave and Compound USDC supply rates with a lag of roughly six hours, by the elasticity-of-substitution mechanism that [31] documents for closed-form market-depth substitution in unified-pool systems. Source: subgraph events, free.

F2 (order-book dynamics). Pending mempool transactions targeting each pool — deposits, withdrawals, borrows, repays — are observable *before* they execute, exactly as a Glosten–Milgrom dealer observes a quote-revision intent before the trade [30, Ch. 3]. This is the most direct analog of the ATD "Goldman leaves its bid" signal that [32] describes on pp. 102–104. We defer F2 to future work because historic Flashbots mempool snapshots have documented gaps in 2024–2025 (design-spec risk R1).

F3 (fragmentation). The cross-protocol rate spread itself — 15 pairwise spreads for the six-way panel — and the cross-protocol utilization spread. This **is** our decision variable, and Section 5.4 confirms it as the dominant contributor to T3's hazard fit, with the synthetic-Cox `f3_spread_max_minus_min` coefficient at $\approx +500$ vs noise-level F1/F4 coefficients. [7] reports a Compound-leads-Aave cointegration with a 0.607 speed-of-adjustment coefficient; our event-time replay shows that this leads-and-lags structure is exploitable at the per-block granularity but not at the hourly aggregation level used in our prior published work [24], which is the empirical resolution of that earlier H_0 verdict.

F4 (related instruments). ETH spot price (drives gas regime), top-LP concentration per pool (more transparent on-chain than in MacKenzie's anonymous order books), and USDC / USDT / DAI peg deviations as leading indicators of liquidity stress; the Aave-vs-Compound cross-chain comparative work of [17] catalogues the relevant risk-parameter levers.

The MacKenzie-to-DeFi mapping is the paper's central *theoretical* contribution and stands independently of the H_1 binding outcomes. The Kissell impact-vs-edge inequality [29] supplies the gating threshold that converts each of F1, F3, and F4 into an actionable trigger.

6.2 The hinge: allocator + protocol launches as mutually reinforcing

Andrew Abbott's notion of a *hinge*, recapitulated by [32, pp. 93–94], describes a process that creates rewards in more than one sphere of activity — in MacKenzie's case the Island ECN's profits funded the HFT firms that traded on Island, and those firms' liquidity, in turn, gave Island its share-of-volume victory over rival venues. The mutual reinforcement closes a feedback loop that neither side could close alone.

The DeFi analog is sharper than the HFT original. A multi-protocol gas-aware allocator's fragmentation-rent (arbitrage across Aave, Morpho, Spark, Compound, Fluid, Euler rate quotes) simultaneously **(a)** earns the allocator alpha, and **(b)** makes it rational for a new lending protocol to launch — because differentiated-rate venues now attract sophisticated flow that bootstraps TVL, which feeds back into the rate-quote signal that the allocator profits from [7]. The empirical evidence is the protocol ordering visible in Table 3: Spark, Morpho Blue, Fluid, and Euler V2 all launched after the 2024-Q4 baseline window, and each is meaningfully on the Pareto frontier of (yield, TVL, switching cost) for at least one of the test-window quarters reported in Table 5. The pre-condition that [32, p. 95] requires for a hinge to close — *unified clearing* — DeFi enjoys for free, because Ethereum L1 is the shared settlement substrate for all six in-scope protocols. Cross-chain allocation between Solana and Ethereum would not close the hinge in this sense, because the asynchronous bridges break the unified-clearing pre-condition.

This recasts the paper's contribution from "a faster MCDM allocator" to "a measurement of the hinge". The Sharpe gap that T1 captures over B4 — +61 bp net APY on the Sep–Dec 2025 validation slice (commit `398809d`), persisting into the test window per Section 5.2 — is not just engineering; it is the rent that any allocator at the per-block resolution can extract from the multi-venue equilibrium, and its sign and magnitude are an observable about the maturity of the DeFi-lending market. The gas-cost-crossover finding of Plan C (at the calibration point of \$17.5 gas per rebalance on a \$1 M position, a switch pays only if the spread is ≥ 5 pp or the position is $\geq 5\times$ scale) quantifies the minimum hinge-rent visible to any participant: below that floor, the allocator does not close the loop and protocol launches do not bootstrap. Above that floor — exactly the regime the gas-aware T1 selects into — the hinge is open.

6.3 Limitations: MEV exposure and the asymmetric-speed-bump fix

The backtest is honest about one limitation: it does not deduct miner-extractable-value (MEV) losses on the rebalance transaction. With public-mempool submission, a \$1 M rebalance would face an expected 5–30 bp sandwich-extraction tax, which would erode much of the T1 edge documented in Table 3. The published agent therefore submits rebalances through the Flashbots private mempool via `eth_sendPrivateTransaction`, as described in Section 3..

We reframe this protection in MacKenzie's terms. [32, pp. 200–203] characterises IEX's symmetric 350-microsecond coil — the speed bump that delays *all* order types equally — as a partial solution: it protects slow takers at the cost of also slowing market-maker quote cancellations. The Flashbots private mempool is structurally different: it delays *visibility* to MEV bots until inclusion — an asymmetric speed bump in MacKenzie's sense — while our own intent-cancellation and re-pricing remain fast. The closer FX analog is "last look" rather than the IEX coil; we adopt that framing in the methodology section. The Plan E live-agent task verifies the Flashbots submission path end-to-end on Sepolia testnet, completing the bridge from backtest to production that the H_1 binding tests presuppose, and the [17] cross-chain comparison establishes the parameter-symmetry conditions under which this generalises.

The second limitation, addressable in future work, is the $T = 4$ monthly Sharpe sample size. With $T = 12$ (one calendar year), the $\sqrt{T-1}$ pre-factor in the Lopez de Prado DSR formula [28, Ch. 14.7.3] would relax by a factor of $\sqrt{11/3} \approx 1.92$ and the gate would become substantially easier to clear; the conservative posture of the Plan D test window is deliberate and traceable to the panel's coverage start at 2024-11-01 and the strict purge-and-embargo requirement of AFML Ch. 7.4. The third limitation is the deferred F2 mempool channel: once Flashbots historic snapshots stabilise, the $X^T X$-style regression of [32, p. 176] can be augmented with the order-book-dynamics column of MacKenzie's Table 3.2, which we expect will close a substantial fraction of the gap to the Kissell information ceiling [29] — specifically the $\sim 40\%$ residual that Section 5.4 attributes to unforecastable adverse-selection noise per [30].

The honest-H_0 posture closes the discussion. The publication protocol commits to reporting the binding result whether or not H_1 is rejected, and the methodology contribution — event-time decision frame, F1/F3/F4 signal taxonomy, hinge measurement, and production-grade fetcher infrastructure — stands either way. The allocator-as-XTX bridge to [32] and the hinge-measurement reframing of the contribution are the durable deliverables; the H1 verdict is an empirical update on the maturity of the multi-protocol equilibrium at the time of the test window.

7. Limitations and Threats to Validity

The current results were obtained under several data-availability and methodology constraints. We list these explicitly so that downstream readers can scope conclusions appropriately and so that reproductions can target the same envelope.

Compound TVL is a placeholder (Finding #3). The Compound V3 RPC fetcher records `total_supplied_usd = 0.0` as a placeholder pending wiring of the USDC base-units → USD conversion. Consequently `tvl_compound`, `debt_compound`, and the derived `dTVL_compound_24h` feature column are identically zero across all 13,105 hourly rows of `joined_clean.parquet`. Branch B of the forecaster therefore receives one constant input ($\approx 14\%$ of its 7-dimensional input capacity), which the model ignores after the first few epochs but which represents wasted capacity.

Ethereum gas price is a constant stub (Finding #4). `data/fetch_gas_eth.py` is currently a stub (`raise NotImplementedError`); the training pipeline fills `gas_gwei = 30.0` as a constant for the entire data window. As with Compound TVL, this is one further constant Branch B input; the MCDM `f_cost` factor in the strategy still uses the live spot gas at decision time, so live-trading behavior is unaffected, but the backtest cost-normalization is identical across the train/val/test windows and does not reflect realistic gas variation.

Aggregate-only metrics, no per-quarter breakdown (Finding #9). The headline metrics in `backtest/run_main.py` (net APY, annual Sharpe, max drawdown, Calmar, forecast hit-rate) are computed over the full test window as a single aggregate. Given the regime structure documented in `CLAUDE.md` (Q4 2024 standard deviation 5.75pp vs 2026 Q1 0.57pp — approximately 10× volatility variation), this aggregation can mask edge concentration in a single regime. A per-quarter breakdown table is queued for the next paper cycle.

Composite-loss components logged in unscaled form (Finding #10). `forecaster/losses.py` logs the bare MSE, $1-\rho_w$, and quantile-loss components to MLflow rather than the scaled $\alpha \cdot \mathrm{MSE}$, $\beta \cdot (1-\rho_w)$, $\gamma \cdot Q_{90}$ contributions.

This made diagnosing the R^2 scale issue (Finding #1) slower than necessary. Scaled-component logging is a low-risk addition deferred to a follow-up.

Tie-break bias in winner-series accounting (Finding #11). `backtest/run_main.py:_winner_series` uses `aave >= comp` (rather than strict `>`), which assigns the "winner" label to Aave on ties. In real data ties occur at machine precision only and the effect on `n_rebalances` is bounded by ± 1; we document this as a convention.

Boundary leakage from 24-hour pct_change feature (Finding #12). `data/features.py` computes `dTVL_aave_24h = tvl_aave.pct_change(24)` on the full joined panel before the train/val/test split is applied, then slices by row index. The first 24 rows of each non-training fold therefore use TVL inputs from the previous fold. The contamination is bounded at $24/2501 \approx 0.96\%$ of test samples; we report it as a threat-to-validity rather than re-engineer the feature pipeline per-fold for this cycle.

***N*-way allocation across the lending market.** Our empirical study covers the two largest Ethereum L1 lending protocols (Aave V3 + Compound V3, jointly 41% of the $\approx$ \$54 B total lending TVL — DeFiLlama April 2026 [18]; see Table 1). The dual-branch architecture generalizes to N protocols without redesign — each additional protocol requires only its protocol-specific kink function $f^{(i)}_{\text{kink}}(u)$ and a per-protocol Branch B feature wiring. Four immediate Ethereum L1 targets, ordered by TVL: Spark (SparkLend, \$6.8 B; Maker-related), Morpho Blue (\$4.9 B; permissionless markets), Fluid (\$1.6 B), and Euler V2 (\$0.89 B). Cross-chain extensions (JustLend on Tron, \$2.4 B; Kamino on Solana, \$1.1 B) defer the rate-fetching infrastructure work but pose no methodological obstacle. We leave the N-way extension as the obvious next step.

8. Conclusion

We tested whether a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) calibrated for LOB mid-price prediction [9] transfers — without architectural change — to DeFi USDC lending-rate forecasting on Aave V3 and Compound V3. Under the same composite loss and the substitution of the domain-natural decomposition pair from (price, volume) to ($\varepsilon = r - f_{\text{kink}}(u)$, $u+\text{ctx}$), the network trains stably on 12,895 hourly bars and attains validation weighted-Pearson 0.747 on Aave with directional accuracy 0.739 — a non-trivial in-distribution signal. The pre-registered H1 test on the binding 4-month test fold, however, returns $\Delta\textsc{Sharpe} = -900.40$ with 95% CI $[-1799.57, -1.23]$ and $p = 1.00$: *fail to reject* H_0. The hysteresis ablation rules out a mis-tuned threshold as the cause; the documented 2025-Q3 $\rightarrow$ Q4 regime inversion is the proximate explanation.

We report the result honestly per the H0-publishable protocol. The contribution stands at three layers. At the architectural layer, identical wiring carries signal across two structurally different financial markets, supporting the design template as portable. At the infrastructure layer, the open-source Compound V3 base-rate loader and the `BaseLendingAllocationStrategy` abstraction upstream to `fractal-defi` [10] are, to our knowledge, the first of their kind. At the empirical layer, the falsified LOB-to-DeFi-lending strategic transfer — mirroring the "negative ensemble" finding of [9] §5.5 one layer up the stack — is itself publishable evidence that complements the equilibrium impossibility result of [11]. Regime- conditional retraining, longer test folds, and N-way protocol extension (Section 7.) are the obvious next steps.

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